

PAVAN AGARWAL

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EDUCATION

University of Wisconsin-Madison, Information School, USA

Aug 2023 - May 2025

Master of Science in Information Systems and Technology [STEM Designated]

(GPA 4/4)

- Data Science Programming (Python), Analytics for Decision Making, Database Design, Data Mining and Machine Learning.

Thadomal Shahani Engineering College, University of Mumbai, India

Jul 2017 - Aug 2021

Bachelor of Engineering in Computer Engineering with Distinction

(GPA 3.8/4)

- Big Data and Analytics, Data Structures and Algorithms, Data Warehousing and Mining, Cloud Computing with Pyspark.

PROFESSIONAL EXPERIENCE

Tesla Inc, Austin TX

Data Engineer - Intern

May 2024 - Present

- Build a standardized system for pipeline development for the Global Quality team, automating pipeline for decision-making for Quality engineers increasing failure **detection by 30%** and reducing downtime by **2 hours**.
- Integrated a Web UI feature using Streamlit Python, helping track cells at the end of the manufacturing process saving **8+ hours** for quality checks also created **6+ tableau visualization** further assisting operations.

Quantiphi Analytics Solution Private Limited, India

Data Engineer - Full-time

Nov 2021 - Aug 2023

1] Development of New Aesthetics Data Pipeline

- Cleaned semi-structured data from 5+ sources in the form of XML, Excel, and structured data directly from on-premise SQL Server to retrieve **1M+** rows of data using SSIS Packages, Python, Spark, and SQL.
- Consulted with the stakeholders to get insights on data and business requirements and revamped data model enhancements to include critical information in the decision-making process.
- Optimized and enhanced Tableau and Power BI dashboards to help stakeholders derive business strategies and helped them **save 4+ hours of manual work per week** and maintained data pipeline uptime to **more than 95% per week** for enhanced efficiency and accuracy.
- Successfully resolved 40+ ad hoc tickets by conducting root cause analysis of production issues and collaborating with business stakeholders to ensure data accuracy, enhancing user experience by 80%.

2] Development of Data API for External vendors (Proof of Concept)

- Architected and designed the API and Generated business for the company by providing a new reporting system, leveraging the API to create PDF reports, assisting and aiding the other team member's smooth transition to the new system, maintaining 90% overall API quality, and reducing programming efforts by 95% during migration.

Quantiphi Analytics Solution Private Limited, India

Data Engineer - Intern

June 2021 - Nov 2021

- Efficiently implemented a data extraction and migration project, leveraging AWS Cloud Redshift that **enhanced data retrieval performance by 200%**, surpassing on-premise datastore capabilities.
- Developed automated ETL (Extract, Transform, and Load) pipeline using distributed computing Python, SSIS Packages, Snowflake, and SQL Script to load Ceded Insurance data daily into the Data Model, **leading to 100% Automation**.
- Streamlined data validation by creating Tableau & Power BI dashboards across 3+ different source systems to substantiate Data Quality checks after loading data into the data model, **saving 2 hours of manual effort daily**.
- Integrated **20GB+** historical data of **200M+** rows into the Data Mart using PySpark and Snowflake for recording past performance and visualized data using Tableau & Power BI for generating insightful reports for the higher management to make business decisions.

EXPERIENTIAL LEARNING EXPERIENCE

Statistical and Exploratory Data Analysis on Air Pollution (USA) | R - Studio, Tableau, Power BI, Python | [Github link](#)

- Presented and Analyzed Air Quality data In the United States by cleaning, merging, and visualizing the parameters, and identified steps to mitigate the primary air pollutants in the air.

Project Management - University Healthcare Vaccination System | MS Office, Presentation | [Github link](#)

- Headed a cross-functional team of four students to assess and perform a thorough analysis of the operational challenges encountered by the University Healthcare Vaccination system. Formulated a comprehensive solution with well-defined objectives, and tangible business value, and identified various stakeholders and potential risks.

RELEVANT SKILLS & CERTIFICATIONS

- **Certifications**- Amazon Web Services (AWS) Certified Solutions Architect Associate, Amazon Web Services (AWS) Certified Developer Associate, Associate in General Insurance Business (AINS - 101), Spark Developer Associate (In progress), Snowflake Certified.
- **Languages** - Spark SSIS, Packages, R, SQL, Python, Pyspark, JavaScript, ShellScript, C, C++, Java, and Distributed Computing.
- **Software** - Tableau, Microsoft Power BI, Microsoft Excel, Visual Studio Code, Microsoft Azure DevOps, JIRA, Git, MySQL, Microsoft SQL Server, Redshift, Oracle, PostgreSQL, Dynamo DB (NoSQL), Dask, Ray, Arrow, Rapids, and Miro.

HONORS AND POSITIONS OF RESPONSIBILITIES

Helper at Data Science Hub | UW-Madison

Jan 2023 - Present

Event Co-ordinator Board Member - Wisconsin International Student Association, UW-Madison | [link](#)

Sep 2023 - Present

Badger Market Cafe Supervisor | UW-Madison

Aug 2023 - Present

Undergraduate Scholarship Award | **100% tuition scholarship**